Dynamic de fluids incompressible

Definition:

elle etudie le movement (ecoulment) des fluide en tenant compte leur force qui genere ce mvt

outil mathématique:

$f(x+\Delta x) = f(x) + \frac{\partial f}{\partial x}\Delta x \qquad \Delta x = dx \to 0$

$\vec{v} = (u_1, u_2, u_3) = \vec{v} = u_i \qquad (x_1, x_2, x_3) = x_i$

$\nabla f = \frac{\partial f}{\partial x_i} \qquad \nabla \vec{v} = \frac{\partial u_i}{\partial x_i}$

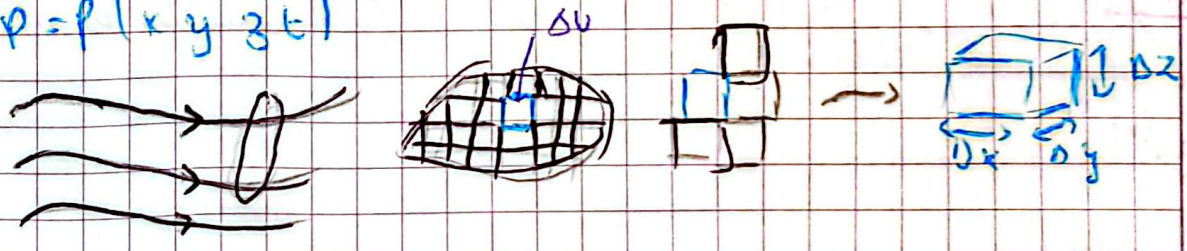
equation de conservation de m ....:

=> objectif : Determination de la vitess et la pression

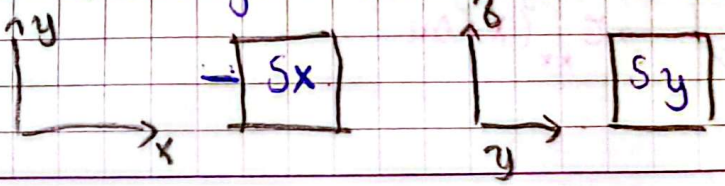
=> procedure de calcule-

soit un fluide reel (f, u), l'ecoulement selon la descreption eulerienne, avec un champ de vitess $\vec{v} = v(x,y,z,t) = (u,v,w$

$\vec{p} = p(x\ y\ z\ t)$



- δv un element d'un fluide
- Les forces sont designé abraves leur direction normal

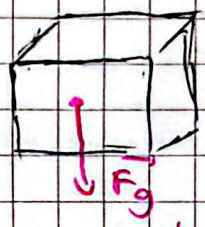


Forces qui agissent sur ΔV :

- Force de volumes :

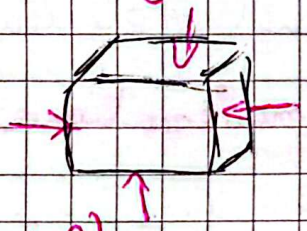
$$\vec{F}_V = m\vec{g} = \rho \Delta V \cdot \vec{g}$$

$$\vec{F}_g = \rho \Delta \vec{g} : g(0,0,-g)$$

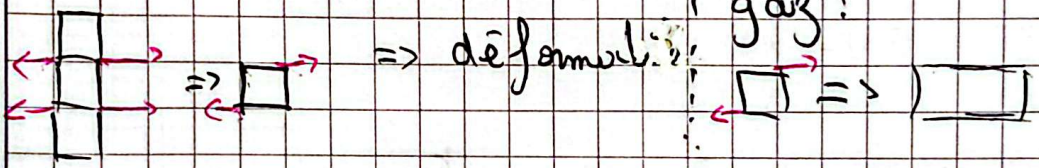


Force de surface :

$$\vec{F}_p = -\nabla p \cdot \Delta V = \frac{-\partial p}{\partial x_i} \Delta V$$

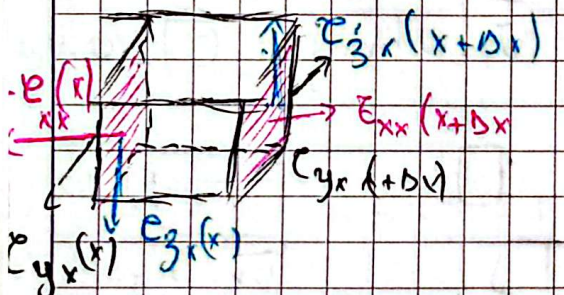


- Force de la viscosité (Force Tensible)



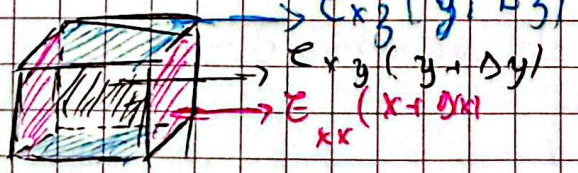
Les force tensible provoque le cisaillement τ et la traction σ

- il sont notée : $\tau(i\,j)$: (la direction normal de la force, la surface)



Les facettes S_x et $S_{x+\Delta x}$

- suivant la direction de x :



Tenseur de contrainte de viles :

$$\tau = \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix}$$

Force lensive :

$$\int \vec{F_i} = \tau_{ij} \cdot S_j = F_{ix}, F_{iy}, F_{iz}$$

. suivant x :

$$\int F_{ix} = -\tau_{xx} \cdot S_{xx} + \tau_{xx}(x+\Delta x) \cdot S_x - \tau_{xy}(y) \cdot S_y + \tau_{xy}(y+\Delta y) S_y$$
$$- \tau_{xz}(z) S_z + \tau_{xz}(z+\Delta z) S_z$$

$$\int F_{ix} = -\tau_{xx} \cdot S_x + \left(\tau_{xx} + \Delta x \frac{\partial \tau_{xx}}{\partial x}\right) \cdot S_x \dots$$
$$= \frac{\partial \tau_{xx}}{\partial x} \Delta x \cdot \Delta y \cdot \Delta z + \frac{\partial \tau_{xy}}{\partial y} \Delta y \Delta x \Delta z \dots$$

$$\int F_{ix} = \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right) \Delta V$$

$$\int F_{iy} = \left(\frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right) \Delta V$$

$$\boxed{\partial \vec{F_i} = \frac{\partial \tau_{ij}}{\partial x_j} \Delta V}$$

$$\varepsilon = u \cdot \partial$$

$$\partial_{ij} = \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}$$

$$\boxed{\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)}$$

$$\delta F_1 = \frac{\partial}{\partial x_j}\left(\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right)\Delta V$$

on suppose que le fluide est newtonien $\mu = cst$

$$\delta F_1 = \mu\left(\frac{\partial^2 u_i}{\partial x_j \partial x_j} + \frac{\partial}{\partial x_i}\left(\frac{\partial u_j}{\partial x_j}\right)\right)$$

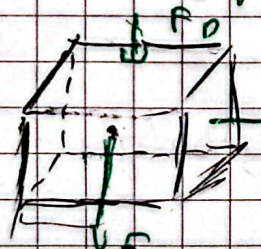
si le fluide est incompressible $\rho = cst$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \text{ou bien} \quad \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3}$$

$$\frac{\partial u_i}{\partial x_i} = 0 \iff \frac{\partial u_j}{\partial x_j} = 0$$

$$\boxed{\delta \vec{F}_1 = \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} \delta V}$$

Bilan des forces



$$\sum \vec{F} = m \cdot \vec{a} \qquad m = \rho \cdot \delta V$$

$$\vec{a} = \frac{D\vec{v}}{dt} = \frac{\partial \vec{v}}{\partial t} + \vec{v} \wedge \vec{v}$$

$$\vec{a} = \frac{\partial v}{\partial t} + u\frac{\partial \vec{v}}{\partial x} + v\frac{\partial \vec{v}}{\partial y} + w\frac{\partial \vec{v}}{\partial z}$$

$$\vec{a} = \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}$$

$$\delta \vec{F}_p = \delta F_1 + \delta F_g = \rho \delta V\left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right) - \frac{\partial p}{\partial x_i}\delta V + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j}\delta V + \rho \delta V g_i = \rho \delta V\left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right)$$

$$\boxed{\rho\left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} + \rho g_i}$$

Forme vectoriel

$$\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v}\,\nabla\vec{v}\right) = -\nabla p + \mu\nabla\vec{v} - \rho\vec{g}$$

Forme algebrique:

ρ |

- Equation

L'equation de quantité de mvt contien 2 inconnus $\vec{v}$ $\vec{p}$ on n'est pas possible de la resoudre sans introduire 2ème equation

equation de continuité

$$\nabla\vec{v} = 0$$

$$\rho\left(\frac{\partial \vec{v}}{\partial t} + v\,\nabla\vec{v}\right) = -\nabla p + \mu\nabla^2\vec{v} + \rho\vec{g}$$

equation de Navier-stokes:

$$\frac{\partial u_i}{\partial x_i} = 0 \qquad \rho\left(\frac{\partial u_i}{\partial x_i} + u_j\frac{\partial u_i}{\partial x_j}\right) = \frac{-\partial p}{\partial x_i} + \mu\frac{\partial^2 u_i}{\partial x_j \partial x_j} + \rho\vec{g}$$

Forme vectoriel:

$$\nabla\cdot\vec{v} = 0$$

$$\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v}\cdot\nabla\vec{v}\right) = -\nabla p + \mu\nabla^2\vec{v} + \rho\vec{g} \qquad g\begin{pmatrix}0\\0\\-g\end{pmatrix}$$

en 3D cartesienne :

$$\rho\left(\frac{du}{dt} + u\frac{du}{dx} + v\frac{du}{dy} + w\frac{du}{dz}\right) = -\frac{dp}{dx} + \mu\left(\frac{d^2u}{dx^2} + \frac{d^2u}{dy^2} + \frac{d^2u}{dz^2}\right)$$

même chose pour v et w

équation d'Euler :

pour le fluid parfait $\mu = 0$ $\quad \nabla \cdot \vec{v} = 0$ $\quad \rho\left(\frac{\partial \vec{v}}{\partial t} + v\,\nabla \vec{v}\right) = \nabla p + \rho g$

Equation de Burgers :

on neglige la pression de fluide

$$\rho\left(\frac{\partial v}{\partial t} + \vec{v}\,\nabla\vec{v}\right) = \mu\,\nabla^2\vec{v} + \rho\vec{g}$$

Conservation d'énergie le long de ligne de courant (Euler)

Supposons : fluide parfait ($\mu = 0$)

Ecoulement stationnaire $\frac{\partial \vec{v}}{\partial t} = 0$

$$\rho \vec{v} \cdot \nabla \vec{v} = -\nabla p + \rho \vec{g}$$

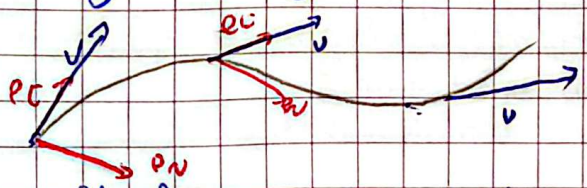
- en analyse vectorielle :

$$\vec{v} \cdot \nabla \vec{v} = \frac{\nabla \vec{v} \cdot \vec{v}}{2} - \vec{v} \wedge (\nabla \wedge \vec{v}) \qquad \vec{\omega}$$

$$\rho \nabla \left(\frac{v \cdot v}{2} \right) - \rho \vec{v} \wedge \vec{\omega} = \nabla(-p) + \nabla(-\rho g z)$$

$$\nabla \left(\frac{\rho v^2}{2} + p + \rho g z \right) = \rho \vec{v} \wedge \vec{\omega} \quad \dots (1)$$

- Le long d'une ligne de courant



$$\vec{e}_C = \frac{\vec{v}}{v}$$

$\|\vec{e}_N\| = \|\vec{e}_C\| = 1$

on multiplie par un produit scalaire l'eq (1) par $\vec{e}_C$

$$\nabla \left(\frac{\rho v^2}{2} + p + \rho g z \right) \cdot \vec{e}_C = \rho (\vec{v} \wedge \vec{\omega}) \cdot \vec{e}_C$$

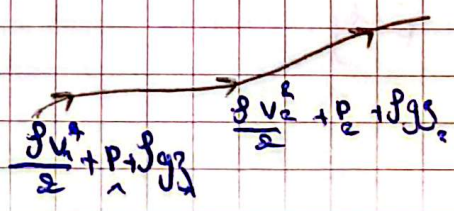
$$\nabla (\quad) \vec{e} = \frac{\rho}{v} (\vec{v} \wedge \vec{\omega}) \cdot \vec{v} = 0$$

$$\nabla (\quad) \vec{e} = 0$$

donc $\nabla \left(\frac{\rho v^2}{2} + p + \rho g z \right) = cst$

Le long d'une ligne de courant

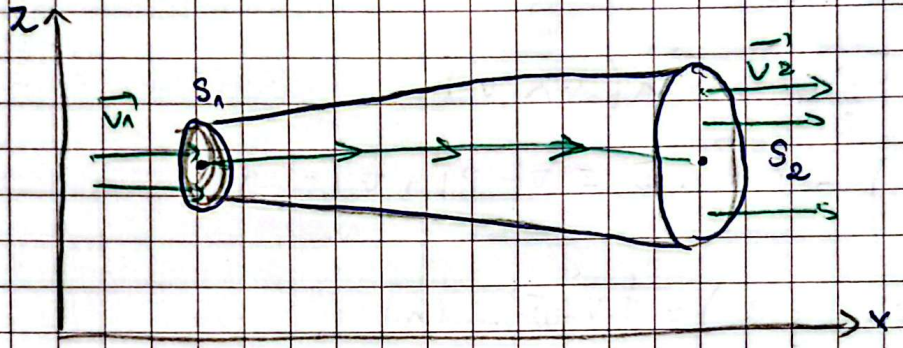
$$\frac{\rho v^2}{2} + p + \rho g z = cst$$



Théorème de Bernoulli :

Sur un tube de courant dont il est possible de le considérer mince, on peut adopter la forme de conservation de $\frac{\rho v^2}{2} + \rho g z + p$ sur ce tube de courant.

- Tube de courant delimité par deux section S_1 et S_2



- un champ de vitesse uniforme sur la section (1)

$$p_1 + \frac{\rho v_1^2}{2} + \rho g z_1 = p_2 + \frac{\rho v_2^2}{2} + \rho g z$$

le long de ligne de courant médiane

$$v_1 S_1 = v_2 S_2$$

$$p_1 + \frac{\rho v^2}{2} + \rho g z = p_2 + \frac{\rho v^2}{2} + \rho g z_2$$

Equation de Bernoulli :

$$p + \frac{\rho v^2}{2} + \rho g v = cst$$

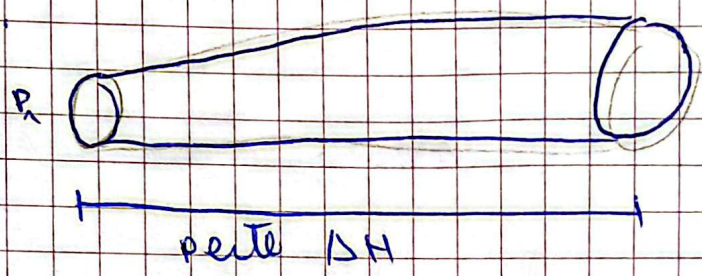
Forme de pression d'énergie

p. statique $\Rightarrow$ p. dynamique

$\frac{P}{\rho g} + \frac{v^2}{2g} + z = cst$ L'Monsieur yKhdm bHaddi

Theorème de Bernoulli généralisé

pour un fluide réel la viscosité de fluide entraine le frottement des particules de fluide avec les parois des conduits avec elle mene cela entraine des pete d'energie qui vont diminuer la pression du fluide ainsi que sa charge : les pertes = ΔH

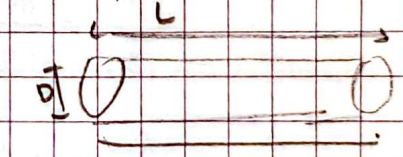


$\frac{P_1}{\rho g} + \frac{\vec{v_1}}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{\vec{v_2}}{2} + z_2$

1/ perte de charge reguliere : ΔH_r

causé par le frottement pour une conduit cylindrique

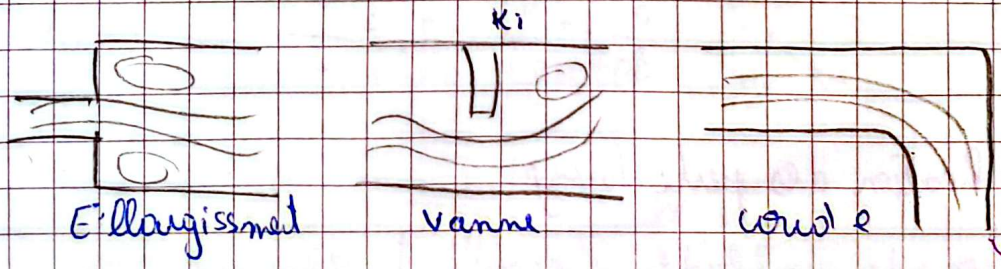
$\Delta H_r = \lambda \frac{L}{D} \frac{v^2}{2g}$



λ = coef de Frot

2/ perte de charge singuliere

causé par l'existance d'obstacle dans les conduites



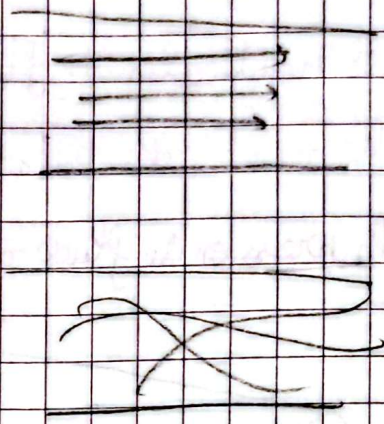
$\Delta H = \Delta H_r + \Delta H_s$ $\quad \Delta H_s = \sum K_i \frac{v^2}{2g}$

coef de construction

Calcule de λ:

Regime d'écoulement.

$$Re = \frac{F\ d'inertie}{F.\ de\ viscosité} \Rightarrow Re = \frac{\rho v D}{\mu} \quad si\ Re < 2000$$



1. Le regime laminaire: $\lambda = \frac{}{Re}$

2) $Re > 2000$ Regime turbilent $Re < 10^5$

$\lambda = 0,196 . Re^{-0,25}$

3) turbilent regueux.

$\lambda = 0,79 \sqrt{\frac{r}{D}}$



Theorème de Bernouli avec echange denergie

Dans une ligne contenant une pompe ou une turbine

pompe

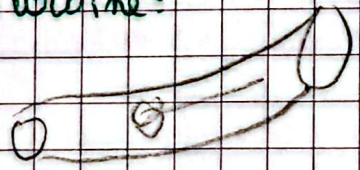


$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + \frac{P_0}{Q_m}$$

$$\frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + \Delta H$$

$Q_m = \rho v S$

Turbine:



$$\frac{P}{\rho g} + \frac{v^2}{2g} + z_1 = \frac{P}{\rho_2} + \frac{v}{2g} + z_2 + \Delta H + \frac{P_0}{Q}$$

Explication des perte charge

on consider un fluide s'ecoulant dans une pipe horizontalle detuite de presometrique

Les tubes sont très long le fluide a l'intérieur de tube est statique

1) on suppose que le fluide soit parfait $\mu = 0$. on aplique T. Ber

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + \Delta H^{\nearrow 0} \qquad Q_V = cst$$

$$V_1 S_1 = V_2 S_2 \qquad S_1 = S_2 \Rightarrow V_1 = V_2 \quad \text{on a} \quad z_1 = z_2$$

LFH : $P_1 = P_{atm} + \rho g h_1$ / $P_2 = P_{atm} + \rho g h_2$

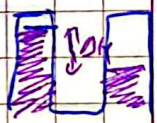
$$\frac{P_{atm} + \rho g h_1}{\rho g} + \frac{v^2}{2g} + z = \frac{P_{atm} + \rho g h_2}{\rho g} + z + \frac{v^2}{2g}$$

$$h_1 = h_2 \Rightarrow P_1 = P_2$$

2) fluide reel: $\mu \neq 0$ chute de pr

$$\frac{P}{\rho g} = \frac{P}{\rho g} + \Delta H \qquad \Delta H = \frac{P_1 - P_2}{\rho g} \Rightarrow \Delta P = \rho g \Delta H$$

$h_1 - h_2 = \Delta H$ perte de charge $\Rightarrow h_1 > h_2$



Tube de venture : convergent divergent



- fluide.

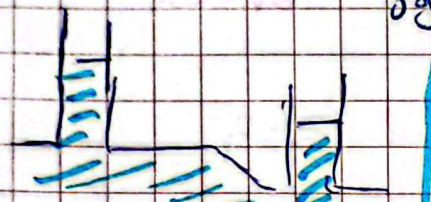
fluide parfait :

Le but est déterminer la chute de pression

Bernoulli entre 1-2 : $\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v^2}{2g} + z_2$

$\frac{S_1}{S_2} > 1 \Rightarrow \boxed{v_2 > v_1}$ $\quad$ $v_1 S_1 = v_2 S_2 = Q_v$ $\quad$ $v_2 = \frac{v_1 S_1}{S_2}$

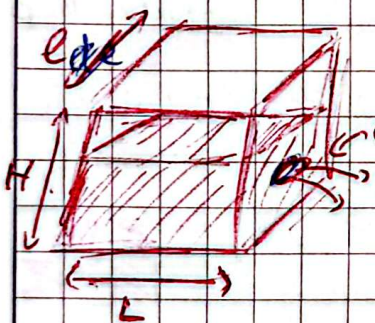
Dautre part : $\frac{P_2 - P_1}{\rho g} = \frac{v_2^2 - v_1^2}{2g}$ $\quad$ $v_2 > v_1 \Rightarrow P_1 - P_2 > 0$ $\quad$ $\boxed{P_2 < P_1}$



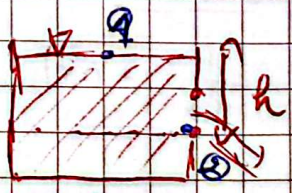
$\boxed{v \nearrow \; P \searrow}$ l'augmentation de la vitesse provoque la diminution de la pression statique et augmente la de la pression dynamique $\frac{\rho v^2}{2}$

Formules torricelli :

L'ecoulement a travers un orifice on cosidere un reservoir



Evacuation vidange jet $\quad Q_v = \text{debit} ?$



fluide parfait Bernoulli 1-2

$\frac{P}{\rho g} + \frac{v_1^2}{2g} + z = \frac{P}{\rho g} + \frac{v^2}{2g} + z$

$v_1 S_1 = v_2 S_2$ $\quad$ $v_1 = \frac{v_2 S_2}{S_1} = \frac{S_2}{e \cdot L} = \frac{}{\infty} = 0$ $\quad$ $v_1 \simeq 0$